

A profusion of neural stem cells in the brain of the spiny mouse, *Acomys cahirinus*

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Abstract

The vast majority of neural stem cell studies have been conducted on the brains of mice and rats, the classical model rodent. Non-model organisms may, however, give us some important insights into how to increase neural stem cell numbers for regenerative purposes and with this in mind we have characterized these cells in the brain of the spiny mouse, *Acomys cahirinus*. This unique mammal is highly regenerative and damaged tissue does not scar or fibrose. We find that there are more than three times as many stem cells in the SVZ and more than 3 times as many proliferating cells compared to the CD-1 outbred strain of lab mouse. These additional cells create thick stem cell regions in the wall of the SVZ and very obvious columns of cells moving into the rostral migratory stream. In the dentate gyrus, there are more than 10 times as many cells proliferating in the sub-granular layer and twice the number of doublecortin expressing neuroblasts. A preliminary analysis of some stem cell niche genes has identified *Sox2*, *Notch1*, *Shh*, and *Noggin* as up-regulated in the SVZ of *Acomys* and *Bmp2* as being down-regulated. The highly increased neural stem cell numbers in *Acomys* may endow this animal with increased regenerative properties in the brain or improved physiological performance important for its survival.

KEYWORDS

Acomys cahirinus, brain, BrdU, doublecortin, Ki67, neural stem cells, proliferation, spiny mouse

1 | INTRODUCTION

The adult mammalian brain contains at least two populations of neural stem/precursor cells (NPCs) which are set aside during development for the function of repair and replacement throughout life (Lim & Alvarez-Bulleya, 2016; Obernier & Alvarez-Bulleya, 2019). One population is present in the ventricular-subventricular zone (SVZ) on the walls of the lateral ventricle. The neuroblasts generated by these NPCs migrate to the olfactory bulbs which are a considerable distance rostrally by chain migration along a course called the rostral migratory stream (RMS). There they become incorporated into the olfactory circuitry as functional interneurons contributing to fine odor discrimination. It is estimated that many thousands of new olfactory neurons are produced each day in the rodent brain.

The second population of NSCs is present in the dentate gyrus of the hippocampus and pass through a series of developmental stages to become incorporated into the hippocampal circuitry as dentate granule neurons (Goncalves et al., 2016). Here they are involved in learning, memory and pattern separation.

Both populations of NSCs have been investigated with the aim of improving the human condition. In the hippocampus, for example, methods to improve their rate of production, rate of differentiation or slowing their rate of loss during ageing are of considerable interest since if this were to translate to humans the implications for learning and memory in elderly populations or in neurological disease could be highly significant. In this regard, several growth factors and signaling molecules such as Wnt, Shh, BMP, and Notch play crucial roles in the promotion of NSC

proliferation and differentiation often with significant improvements in cognitive function following alteration in their endogenous levels (Ehm et al., 2010; Lai et al., 2003; Yousef et al., 2015). Even relatively simple interferences such as running activity can prevent the decline in hippocampal neurogenesis that occurs with age and improve learning tasks (van Praag et al., 2005).

Subventricular zone stem cell production has been investigated within the concept of regenerative neurogenesis whose ultimate aim is to exploit these endogenous cells for use in repair mechanisms following physical injury or stroke (Obernier et al., 2014). Although neuroblasts have been demonstrated to leave the SVZ or RMS and migrate to the damage site over a prolonged period, they may not survive long term or give rise to the appropriate mature neuronal phenotype (Cui et al., 2013; Liu et al., 2009; Osman et al., 2011) thus restricting their usefulness for regenerative medicine therapies. However, enhancing SVZ stem cell proliferation, migration and differentiation with growth factors has a profound effect of recovery from stroke (Kolb et al., 2007) suggesting that increasing NSC numbers may indeed be a successful strategy for repairing brain damage.

In support of this concept that 'more is better' for regeneration, a comparative view of regenerative neurogenesis (Alluni & Bally-Cuiff, 2016; Ferretti, 2011) reveals that there is a large amount of endogenous NSC proliferation in lower vertebrates such as fish, salamanders and reptiles and the number of cells in the brain increases with body size, length and age (Berg et al., 2010; Ferretti, 2011; Kazlin et al., 2008; Kizil et al., 2012; Zupanc, 2008). In axolotls and newts proliferation is not restricted to specific regions such as the mammalian SVZ but is present throughout the forebrain endowing these regions with extensive regenerative powers (Amamoto et al., 2016; Berg et al., 2010; Maden et al., 2013). In zebrafish there are 16 distinct proliferating domains throughout the anteroposterior axis of the zebrafish brain (Grandel et al., 2006) which similarly enables rapid and efficient neuronal regeneration after brain injuries (Kaslin et al., 2008).

Compared to these regenerating species, mammals have not only restricted their neurogenesis regions to the SVZ and the DG, but the rate of production of new cells is much lower. Because of its importance in learning and memory most comparative studies of adult neurogenesis rates has been performed on the DG where the lowest level of expression is seen in humans, nonhuman primates, bats, cetaceans, shrews and moles (Amrein, 2015; Lipp & Bonfanti, 2016). Rodents appear to have higher levels of adult neurogenesis than other mammals although the variation in levels may be due to species-specific demands that characterize their niche, but since mammals are generally considered to be non-regenerating organisms adult neurogenesis has not been examined with this in mind.

Another factor limiting the potential of endogenous neural stem cells to repair sites of damage is the robust inflammatory response that takes place after brain injury which is in part responsible for the lack of survival of any endogenous stem cells that do arrive at the injury site (Anrather & Iadecola, 2016; Cui et al., 2013; Zhang et al., 2019). With these two concepts in mind: mammals have very limited powers of regeneration, and inflammation potentially reduces

the ability of the brain to undergo self-repair, we have examined the presence of NSCs in the brain of the spiny mouse, *Acomys cahirinus* because this organism displays very unusual, perhaps unique behavior. First, it has striking powers of tissue regeneration in response to tissue damage. The skin, ear punches, skeletal muscle and the kidney readily recover from the damage that is applied to them whether it be tissue removal, burns, myotoxin or ischemia (Brant et al., 2019; Gawriluk et al., 2016; Maden, 2018; Maden et al., 2018; Okamura et al., 2018; Seifert et al., 2012). As a result all the components of the skin such as hair, sebaceous glands, skeletal muscle and the cartilage of the ear can regenerate. Second, in doing so this mammal displays a greatly reduced inflammatory response reflected in decreased numbers of the pro-inflammatory M1 macrophages and decreased cytokine response (Brant et al., 2015, 2016; Okamura et al., 2018; Simkin et al., 2017; Streeter et al., 2020) both of which may be permissive for regeneration. With these two factors in mind, we have examined here neurogenesis in the *Acomys* brain as a prelude to investigating the regenerative properties of this organ. We show that compared to the CD-1 outbred strain of the lab mouse there is a more than 3-fold increase in NPC cell numbers and proliferating cells throughout the normal *Acomys* SVZ which gives rise to a larger RMS. Similarly in the other major neural stem cell region of the brain, the dentate gyrus, *Acomys* shows a 6- to 12-fold increase in proliferating cells and a doubling of doublecortin-positive neuroblasts. This greatly increased cell number and increased proliferation may have significance for the improved performance of several neural systems in *Acomys*.

2 | MATERIALS AND METHODS

2.1 | Animals

Five male or female *Acomys cahirinus* were obtained from our breeding colony housed at University of Florida and were aged from 4 to 6 months of age. *Mus musculus* of the outbred CD-1 strain were purchased from Charles River and a similarly sized mixed sex groups of animals used for the experiments of age 2–3 months old. Four wild *Mus musculus domesticus* were also used as a comparison to these 'domesticated' specimens. The weight of each animal used for this study was recorded after euthanasia as were their brain weights. BrdU was administered intraperitoneally at a concentration of 50 mg/kg dissolved in PBS. Three injections 24 h apart were administered and then the brains were fixed 24 h later.

2.2 | Histology and immunocytochemistry

Brains were carefully removed from euthanized animals and weighed before placing in 4% para-formaldehyde for fixation. Each brain was cut in the mid-sagittal plane and processed for wax histology separately. Each half brain was sectioned at 10 µm and every section counted and mounted on slides so that the precise location of each

section from the midline was known. Sections were dewaxed and stained with Nissl stain according to standard protocols.

For immunohistochemistry dewaxed sections were treated with 1% hydrogen peroxide in dd water to remove endogenous peroxidases. Subsequent antigen retrieval was by microwave heating for 4 min in citrate buffer pH6 (for Ki67 and DCX) or 5 min at high pressure in a pressure cooker containing citrate buffer (for BrdU). The following antibodies were used: BrdU (Accurate Chemical Corp at 1 in 1000), Ki67 (Abcam ab15580 at 1 in 400), PCNA (Dako M0879 clone PC10 at 1 in 500) doublecortin (Abcam ab18723 at 1 in 500). Visualization of antibody reactivity was performed with the Vectorstain Elite kit with a 1 in 400 concentration of biotinylated secondary antibody, ABC reagent and DAB according to the manufacturer's protocol. Some sections were lightly counterstained with hematoxylin and mounted with Cytoseal.

For counting cells, the total number of cells and the number of immunopositive cells were recorded in a 100× field of view on an Olympus compound microscope and at least 20 consecutive sections were counted and the counts repeated on three individual brains. Since every section was collected the precise location of the sections relative to the mid-sagittal plane was known and the counts were performed at similar anatomical locations in each brain. The statistical significances of the counts obtained were determined with the Student's *T*-test and probabilities elaborated in the figure legends.

2.3 | RT-qPCR

For PCR analysis, three brains of *Mus* and *Acomys* were removed from the euthanized animals, cut in the mid-sagittal plane and samples of tissue placed into 1 ml of RNALater. The samples prepared were a 1-mm cubed piece of tissue from the dorsal cortex and for a NSC preparation the tissue on the surface of the caudate putamen was scraped off with a sterile spatula. Samples were stored at 4°C for 24 h and then maintained at −80°C. Samples were subsequently thawed at 4°C and homogenized in RLT Buffer and processed using the RNeasy Plus Mini Kit (Qiagen). RNA quality was assessed using an Agilent 2200 TapeStation to ensure that all samples had a RIN score >7.0. cDNA was generated with SuperScript™ IV VILO Reverse Transcriptase following the manufacturer's protocol. Real-Time PCR was performed using Sso-Fast™ EvaGreen® Supermix (Bio-Rad) on a Bio-Rad C1000 Touch™ Thermal Cycler using species-specific primers. The fold change in gene expression was calculated using the $\Delta\Delta C_t$ relative expression method (Livak & Schmittgen, 2001) using *Gapdh* as the reference gene. The sequence of species-specific PCR primers are recorded in Table S1. All reactions were run with an annealing temperature of 60°C.

3 | RESULTS

We first sought to determine whether there were any differences in brain weights, body weights, or brain:body weight ratios which could

be relevant for any differences which might appear in cell numbers or tissue structure in our animal comparisons. *Acomys* adults are larger than CD-1 *Mus* adults and the animals used here weighed 36% more (*Acomys* 52.77 g; *Mus* 38.88 g, $p = 0.02$; Figure 1a). Perhaps as a result of a larger body size *Acomys* had 50% larger brain weights (*Acomys* 0.775 g; *Mus* 0.509 g, $p = 1.2 \times 10^{-4}$; Figure 1b). However, the brain size as a percentage of body size was 1.4% in *Acomys* and 1.3% in *Mus*, a very similar figure, so any differences found in cell number or tissue structure was unlikely to be due to the larger brains in *Acomys*. To provide further relevant data we obtained several specimens of wild *Mus* (*Mus musculus domesticus*) and these animals had by far the smallest bodies (14.4 g; Figure 1a) than either CD-1 or *Acomys*, but the largest brains of all (1.085 g; Figure 1b). As a percentage of body size the wild *Mus* brain was 7.5%, about 5–6× the figure for *Acomys* and CD-1.

3.1 | The SVZ and RMS

The CD-1 *Mus* (hereafter referred to as *Mus*) and *Acomys* brains were initially examined for superficial anatomical differences and then serially sectioned for more detailed studies. The anatomical view of the mid-sagittal brain of *Mus* and *Acomys* showed expected similarity in the major brain regions with one consistent difference. Instead of the smooth fiber tracts on the underside of the corpus callosum which form the dorsal roof of the lateral ventricle that is seen in *Mus* (Figure 2a), there was what seemed to be an outgrowth which protruded into the lateral ventricle (Figure 2b, red arrow). Sections through this region in *Acomys* stained with Nissl stain showed that this area was packed with a large number of darker staining cells accumulating between the corpus callosum and the caudate putamen (*cp*) within the lateral ventricle (Figure 2e). The horizontal band of cells forms a thick SVZ region running across the *cp* and the large ventral tract of cells is filling up the space between the *cp* and the lateral septal nucleus (*ls*). The equivalent location in the *Mus* brain (Figure 2c) shows a far smaller tract both ventrally between the *cp* and the *ls* and running across the *cp*. In addition to the ventral tract between the *cp* and the *ls* the other classical regions of NSCs are thicker in *Acomys*, namely the caudal wall of the *cp* (Figure 2d *Mus*; Figure 2f *Acomys*) and the horizontal stream across the *cp* (Figure 2g *Mus*; Figure 2i *Acomys*). The thicker horizontal band of NPCs can also be clearly seen in the BrdU-labeled sections in Figure 2h (*Mus*) and Figure 2j (*Acomys*) and in doublecortin-stained sections in Figure 3d (*Mus*) and Figure 3d' (*Acomys*). Measurements of the thickness of this horizontal band of NPCs in serial sections of this region show that the *Acomys* band is 2.4-fold thicker (Figure 1j, $p = 1.2 \times 10^{-4}$).

To confirm this impression of greater numbers and to examine proliferation in the NSCs, cell counts and immunocytochemistry were performed using Ki67, PCNA, and BrdU-labeled sections. A minimum of 12 sections were counted on three brains giving total cell numbers of up to 15,000 cells. The regions counted were the vertical lining of the caudal wall of the *cp* (referred to as

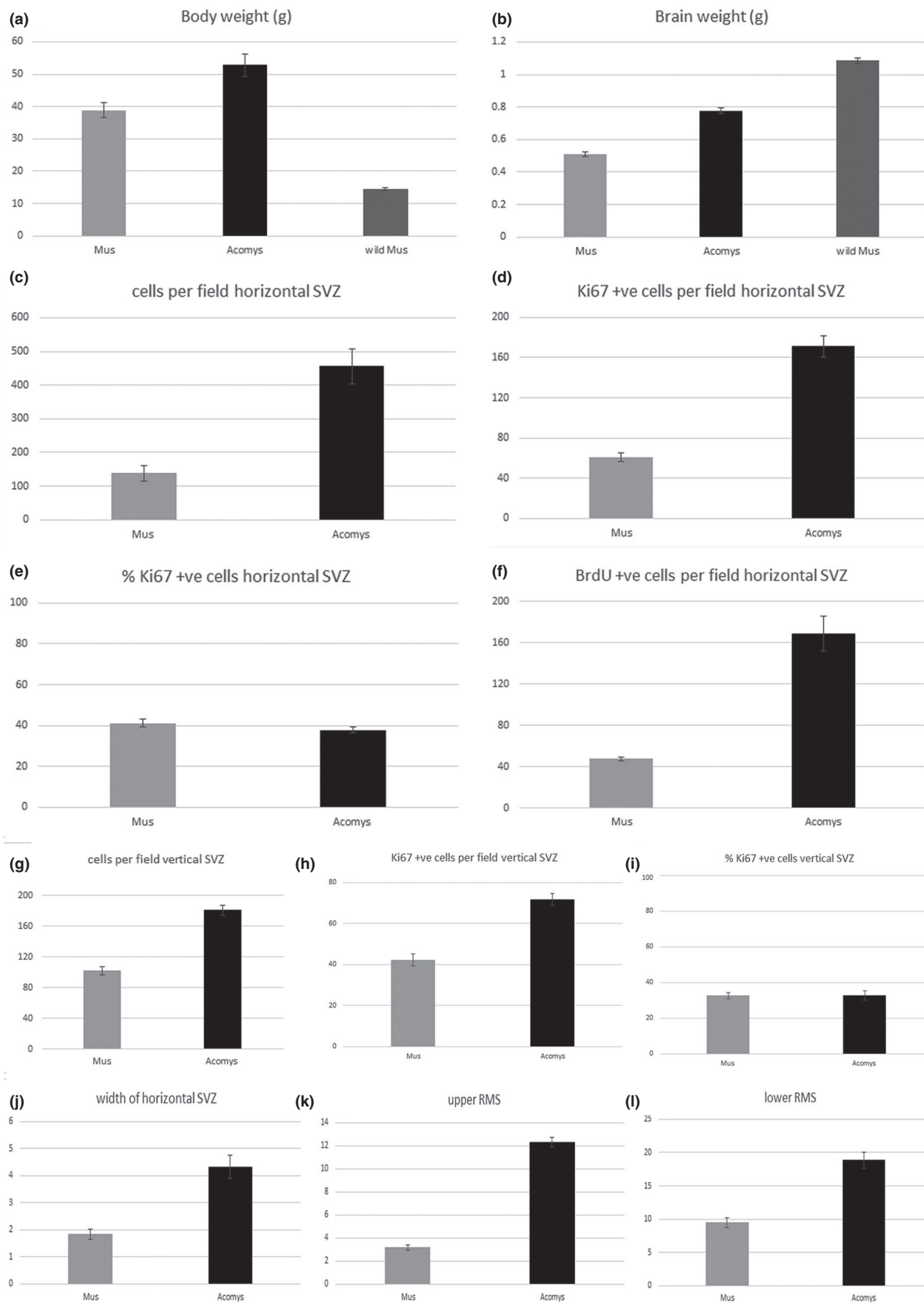


FIGURE 1 Body weights, brain weights, and subventricular zone (SVZ) cell counts of the animals used in this study. (a) Body weights. *Acomys* v *Musp* = 0.02. *Mus* v wild *Mus* $p = 3.4 \times 10^{-4}$. *Acomys* v wild *Mus* $p = 2.1 \times 10^{-4}$. (b) Brain weights. *Acomys* v *Musp* = 1.2×10^{-4} . Wild *Mus* v *Musp* = 1.7×10^{-7} . Wild *Mus* v *Acomys* $p = 5.4 \times 10^{-6}$. (c) Cells numbers in the horizontal SVZ, $p = 1.24 \times 10^{-4}$. (d) Ki67+ve cells in the horizontal SVZ, $p = 9.06 \times 10^{-10}$. (e) % cells positive for Ki67 in the horizontal SVZ, $p = 0.24$, ns. (f) BrdU+ve cells in the horizontal SVZ, $p = 2.9 \times 10^{-6}$. (g) Cells numbers in the vertical SVZ, $p = 2 \times 10^{-5}$. (h) Ki67+ve cells in the vertical SVZ, $p = 2.71 \times 10^{-7}$. (i) % cells positive for Ki67 in the vertical SVZ, $p = 0.98$, ns. (j) Measurements of width of the horizontal SVZ, $p = 1.2 \times 10^{-4}$. (k) Measurements of the width of the upper RMS stream, $p = 1.5 \times 10^{-9}$. (l) Measurements of the lower SVZ stream as it enters the olfactory bulb, $p = 1.3 \times 10^{-5}$. Error bars are $\pm$ SEM

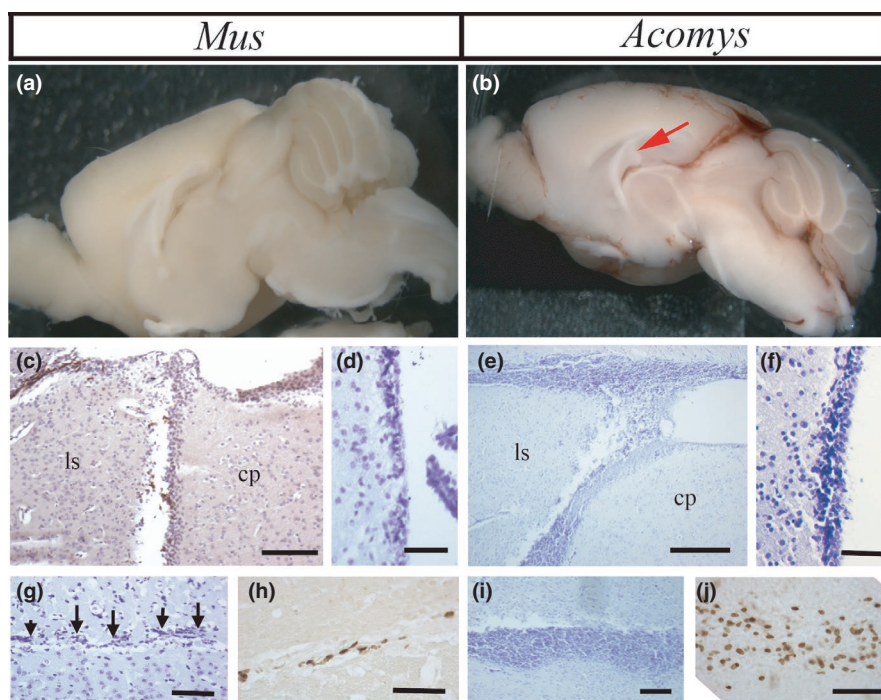


FIGURE 2 Sections of *Mus* and *Acomys* subventricular zone (SVZ) regions. Mid-sagittal view of the brain of *Mus* (a) and *Acomys* (b). Red arrow marks the location of an *Acomys*-specific accumulation of NSCs shown in (g). (c) The only region in the *Mus* brain that resembles any accumulation of NSCs in a T-shape and this is at the rostral edge of the caudate putamen (cp) as it separates from the lateral septal nucleus (ls). Bar = 200 μ m. (d) Nissl-stained NSCs in the vertical caudal wall of the caudate putamen in *Mus*. Bar = 20 μ m. (e) The dense NSC region seen in *Acomys* at the rostral end of the caudate putamen (cp) as it separates from the lateral septal nucleus (ls). Nissl stain. Bar = 200 μ m. (f) Nissl-stained NSCs in the vertical caudal wall of the caudate putamen in *Acomys*. Bar = 20 μ m. (g) The horizontal SVZ region of *Mus* marked by arrows. Nissl stain. Bar = 50 μ m. (h) The horizontal SVZ region of *Mus* labeled with BrdU. Bar = 20 μ m. (i) The horizontal SVZ region of *Acomys* stained with Nissl stain showing a vastly increased number of cells compared to (g). Bar = 100 μ m. (j) The horizontal SVZ region of *Acomys* labeled with BrdU showing a vastly increased number of cells compared to (h). Bar = 20 μ m

vertical SVZ) and the dorsal surface of the cp (referred to as horizontal SVZ). With regard to the horizontal SVZ (Figure 2g,h, *Mus*; Figure 2i,j, *Acomys*) there were 3.3-fold more cells in this location in *Acomys* compared to *Mus* (Figure 1c) with average values of cells per field of 137 in *Mus* and 456 in *Acomys* ($p = 1.24 \times 10^{-4}$). This difference was also seen when the total numbers of Ki67+ve cells were counted in these two same regions and in this case a value of 2.8-fold more cells was obtained (Figure 1d, $p = 9.06 \times 10^{-10}$). Interestingly, when these data were calculated as the % cells proliferating then the same figure of approximately 40% was obtained: *Mus* = 41%; *Acomys* = 37.7% (Figure 1e, $p = 0.24$, ns) suggesting that there may be similar controls in these two populations over cell division. In addition, BrdU labelling was also utilized to add to this proliferation data and counts of the horizontal SVZ gave values of 47 BrdU+ve cells per field for *Mus* (e.g. Figure 2h) and 169 cells

for *Acomys* (e.g. Figure 2j), that is a 3.6-fold higher number of cells in *Acomys* (Figure 1f, $p = 2.9 \times 10^{-6}$), corroborating the Ki67 data.

With regard to the vertical SVZ (Figure 2d, *Mus*; Figure 2f, *Acomys*), there were smaller, but consistent and significant differences in cell number between *Mus* (average value of 111 cells per field) and *Acomys* (average cells per field = 189) giving a value of 1.7-fold greater for *Acomys* (Figure 1g, $p = 2 \times 10^{-5}$). Absolute numbers of proliferating cells measured by Ki67 gave the same significant difference of 1.7-fold (Figure 1h, $p = 2.71 \times 10^{-7}$). As in the case of the horizontal SVZ when this data was calculated as the % cells proliferating in the vertical SVZ identical counts were obtained: *Mus* = 32.6%; *Acomys* = 32.6% (Figure 1i, $p = 0.98$).

The increased numbers of NPCs in these horizontal and vertical SVZ regions generate a larger RMS as might be expected. Measurements of the width of the descending RMS in serial sections

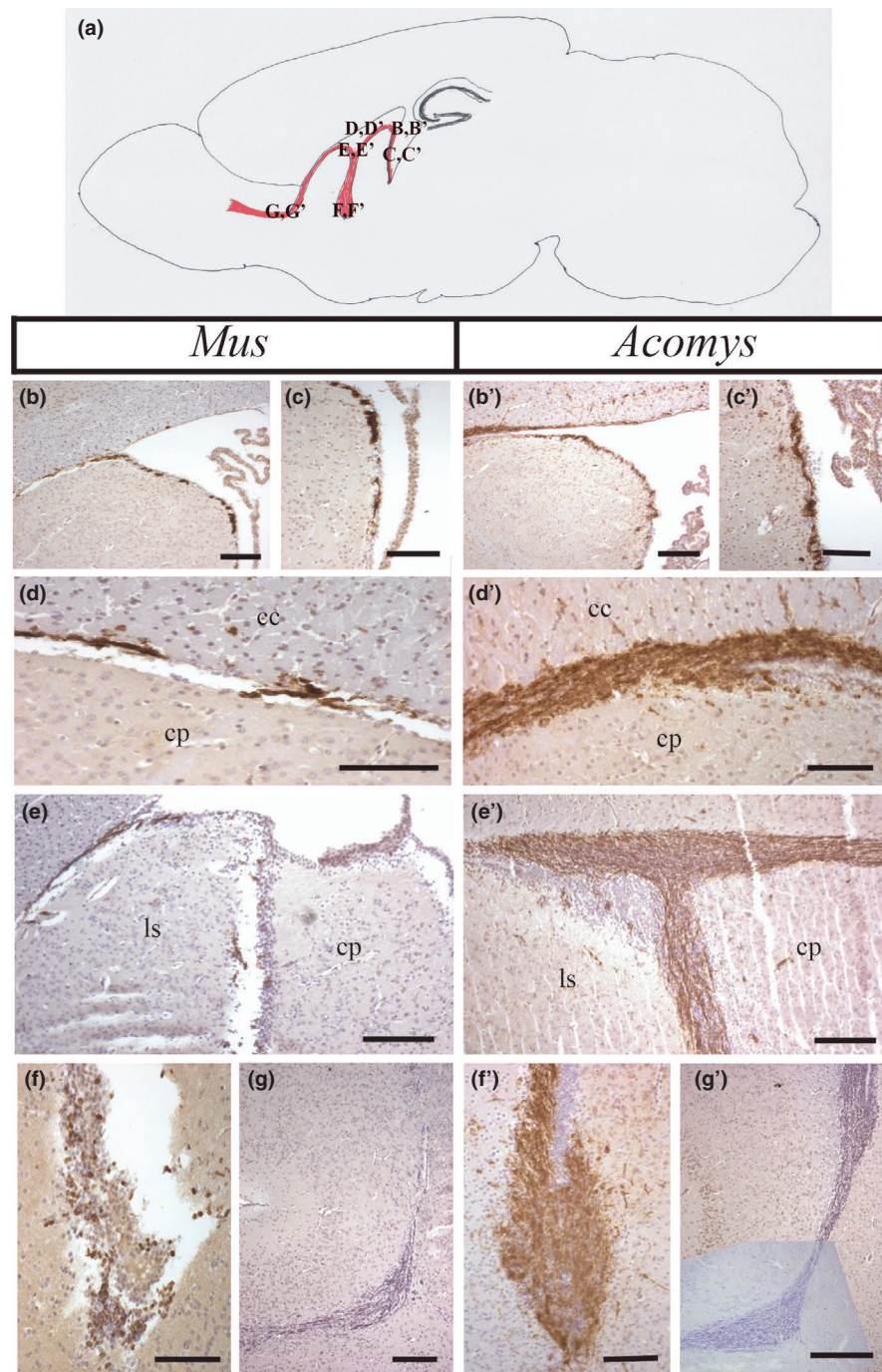


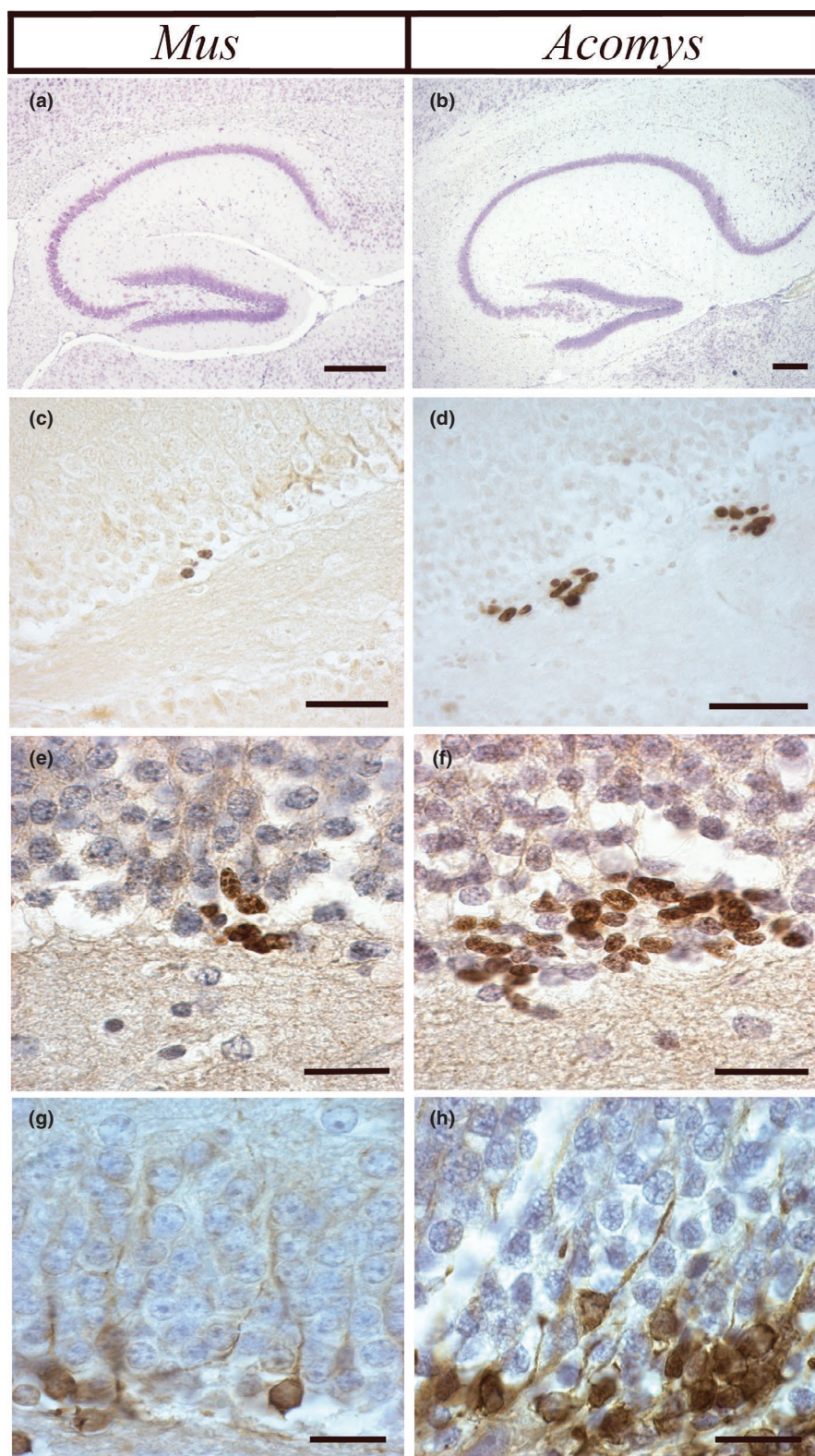
FIGURE 3 Doublecortin-immunostained sections of *Mus* and *Acomys* subventricular zone (SVZ) regions. (a) Drawing of the mid-sagittal brain with the regions of NSCs shown in red. The letters mark the regions that are used for comparison in the subsequent images and these letters correspond to the figure numbers. (b, b') Caudal edge of the caudate putamen in *Mus* (b) and *Acomys* (b'). Bar = 200 μ m. (c, c') Vertical wall of the caudate putamen in *Mus* (c) and *Acomys* (c'). Bar = 100 μ m. (d, d') Horizontal SVZ between the caudate putamen (cp) and corpus callosum (cc) in *Mus* (d) and *Acomys* (d'). Bar = 100 μ m. (e, e') The region of NSCs between the caudate putamen (cp) and the lateral septal nucleus (ls) in *Mus* (e) which has NSCs in the vertical wall and in *Acomys* (e') which is densely packed with huge numbers of NSCs both vertically and transversely. (e) Immunostained with Ki67, (e') immunostained with DCX. Bar = 200 μ m. (f, f') The base of the trough between the lateral septal nucleus and the caudal putamen showing reasonable numbers of Ki67+ve cells in *Mus* (f) and huge numbers of DCX stained cells in *Acomys* (f'). Bar = 100 μ m. (g, g') The RMS in *Mus* (g) and *Acomys* (g') showing greater numbers in *Acomys*. H&E stained sections, g' is a composite photo from two images. Bar = 200 μ m. Cc, corpus callosum; cp, caudate putamen; ls, lateral septal nucleus

show that the *Acomys* stream is 4 $\times$ wider than that of *Mus* (Figure 1k, $p = 1.5 \times 10^{-9}$). Figure 3g (*Mus*) and Figure 3g' (*Acomys*) show hematoxylin-stained sections through the descending RMS and where the RMS enters the olfactory bulb. Measurements of this lower RMS region show a 2-fold increased width in *Acomys* (Figure 1l, $p = 1.3 \times 10^{-5}$). Thus, in all these regions it is clear that there are many more NSCs in the *Acomys* sections than in the *Mus* sections and that the streams of cells in *Acomys* are thicker.

These proliferating cells described above are the transit amplifying or C cells of the NSC niche. To identify the other populations of NSCs, the A cells (type I neuroblasts) and B cells (quiescent NSCs) present in the SVZ we used doublecortin (DCX) and GFAP staining,

respectively. The same results were obtained – the total number of immunopositive cells were far more in *Acomys* than in *Mus* but the proportions were the same. For example, Figure 3 shows DCX immunostaining in the various regions of the SVZ that are identified in Figure 3a. Figure 3b (*Mus*) and Figure 3b' (*Acomys*) show DCX+ve neuroblasts at the caudal edge of the cp. Figure 3c and c' (*Acomys*) show DCX+ve neuroblasts in the vertical wall of the cp (see also Figure 2d, *Mus* and Figure 2f, *Acomys*). Figure 3d (*Mus*) and Figure 3d' (*Acomys*) show DCX neuroblasts in the horizontal stream over the cp with a greatly increased number in *Acomys*. Figure 3e (*Mus*) and Figure 3e' (*Acomys*) show the region between the cp and the ls with a vertical accumulation of neuroblasts which intersects

FIGURE 4 Structure, cell proliferation and neuroblast differentiation in the dentate gyrus (DG) of *Mus* and *Acomys*. (a) Nissl staining of a sagittal section through the hippocampus of *Mus*. Bar = 250 μ m. (b) Nissl staining of a sagittal section through the hippocampus of *Acomys*. Bar = 250 μ m. (c) BrdU+ve cells in the subgranular zone in a section of *Mus*. Bar = 50 μ m. (d) more BrdU+ve cells in the subgranular zone in *Acomys*, with the cells arranged in groups. Bar = 50 μ m. (e) Ki67+ve cells in the subgranular zone in a section of *Mus*. Bar = 20 μ m. (f) more Ki67+ve cells in the subgranular zone in *Acomys*. Bar = 20 μ m. (g) DCX+ve cells in the subgranular zone of a section of *Mus*. Bar = 20 μ m. (h) more DCX+ve cells in the subgranular zone in *Acomys*. Bar = 20 μ m



with the horizontal stream, again with far higher absolute numbers in *Acomys* and Figure 3f (*Mus*) and Figure 3f' (*Acomys*) show neuroblasts at the base of this vertical groove. The same principles were seen with GFAP immunostaining for quiescent NSCs – more total cells in *Acomys* but the same proportions of stained vs unstained cells as in *Mus*.

3.2 | Dentate gyrus

Since *Acomys* has far more NSCs of all types in the SVZ regions and RMS than *Mus* we also looked at another site of NSC production, the dentate gyrus (DG) of the hippocampus. In contrast to the lateral ventricle regions described above there was no obvious

histological difference between the two species in the DG with the pyramidal cell layer, hilus and granule cell layer apparently indistinguishable except for a slightly larger organ size in *Acomys* (Figure 4a,b). However, after BrdU analysis it was obvious that there were far many more BrdU+ve cells in the subgranular zone in a typical section of *Acomys* (Figure 4d) than in *Mus* (Figure 4c). The average number of BrdU+ve cells per section in *Mus* was 6.2 and in *Acomys*, 34.9 (Figure 5a), that is 5–6 fold more ($p = 1 \times 10^{-11}$). To confirm that this was related to proliferation rather than other reasons why cells would incorporate BrdU we used two additional proliferation markers. Using Ki67 a similar result was obtained (Figure 4e,f) although the difference was greater as the average number of Ki67+ve cells per section in *Mus* was 4.4 and in *Acomys* 51.6, that is 11–12 fold more (Figure 5b, $p = 1 \times 10^{-10}$). As has been seen before these proliferating cells appeared in nests throughout the subgranular zone rather than evenly distributed and in *Mus* a typical group size was 2–6 (Figure 4e) whereas in *Acomys* it was 20–30 (Figure 4f). In a third analysis of proliferation the PCNA antibody was used and in this case the results were almost the same as Ki67 as there were an average of 5.2 cells per DG section in *Mus* and 54.1 in *Acomys*, that is 10–11 fold more (Figure 5c, $p = 4 \times 10^{-11}$). All three markers of proliferation therefore suggest that *Acomys* NSCs in the subgranular zone there are many more proliferating cells, perhaps as high as 10-fold, than the equivalent cells in *Mus*.

To determine whether these proliferating cells were differentiating into neuroblasts or being discarded we stained sections with a doublecortin (DCX) antibody to identify immature neurons. Such cells could readily be identified in the subgranular zone of *Mus* with their axons extending up through the granule cell layer (Figure 4g) and the average number of cells per section was found to be 18. Once again there were more DCX+ve cells in an equivalent section of *Acomys* (Figure 4h) with counts revealing an average number of 32, that is 1.8-fold more than *Mus* (Figure 5d, $p = 7.4 \times 10^{-8}$).

3.3 | Some potential gene expression differences in the SVZ

To investigate potential molecular differences which might correlate with these greater cell numbers in the SVZ of *Acomys* compared to *Mus* we performed RT-qPCR with several genes that have been suggested to play a role in the creation of the NSC niche. There may be higher levels of positive regulatory niche genes in *Acomys* or there may be higher levels of negative regulators in *Mus*. There may also be unique genes in *Acomys*, but this would require a more thorough analysis than that performed here. We selected *Sox2*, *Notch 1*, *Notch2*, *Dlx2*, *Shh*, *Noggin*, *Bmp2*, *Gdnf*, and *GFAP* for analysis (Ehm et al., 2010; Lai et al., 2003; Yousef et al., 2015), used species-specific primers to overcome any lack of specificity (Table S1) and

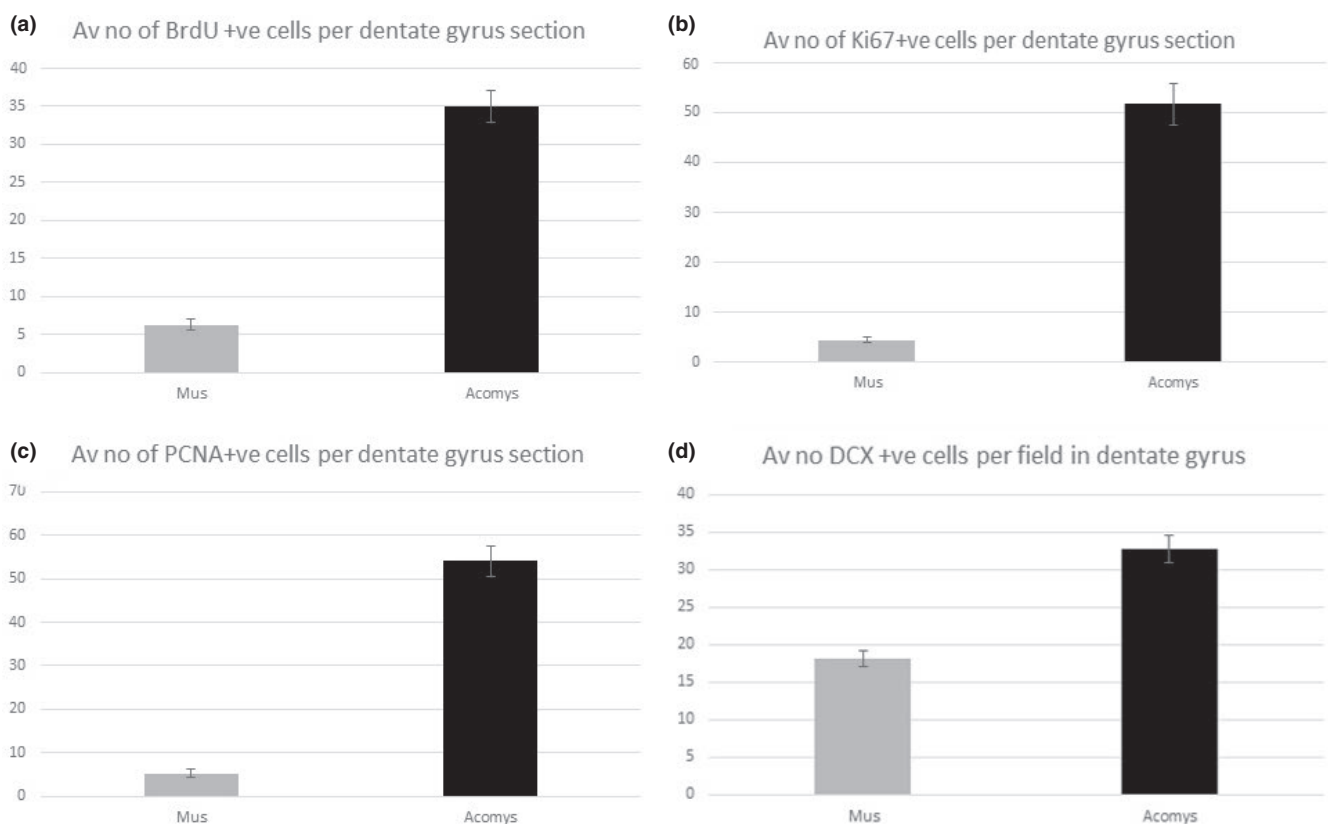


FIGURE 5 Cell counts in *Mus* and *Acomys* dentate gyrus. (a) Average number of BrdU+ve cells, $p = 1 \times 10^{-11}$. (b) Average number of Ki67+ve cells, $p = 1 \times 10^{-10}$. (c) Average number of PCNA+ve cells, $p = 4 \times 10^{-11}$. (d) Average number of DCX+ve cells, $p = 7.4 \times 10^{-8}$. Error bars are $\pm$ SEM

measured the gene expression levels in both the SVZ and the cortex as a control.

Directly comparing individual gene expression levels relative to *Gapdh* between species may be subject to a variety of artefacts so to control for this problem we first asked whether each gene was enriched in the SVZ compared to the cortex (as might be expected for SVZ niche genes) and secondly whether the SVZ enrichment for each gene was greater in *Acomys* than in *Mus*.

The results are shown in Figure 6 and reveal that in *Mus* (grey columns) every gene is up-regulated in the SVZ compared to the cortex between 2 and 10-fold. In *Acomys* (black columns) each gene except *Bmp2* is also up-regulated in the SVZ compared to the cortex between 2 and 8-fold. Uniquely, *Bmp2* is down-regulated in the SVZ compared to the cortex in *Acomys*.

Comparing the two species we see that *Sox2*, *Notch1*, *Shh* and *Noggin* are significantly more highly up-regulated in *Acomys* (relative to the cortex) than in *Mus* suggesting that these niche genes may play some role in the more abundant NPCs found in *Acomys*. Uniquely, *Bmp2* is down-regulated in the *Acomys* SVZ compared the cortex whereas in *Mus* it follows the standard pattern of being

enriched in the SVZ. This may suggest the possible involvement of this gene as a negative regulator of NPCs production in *Acomys*.

4 | DISCUSSION

Adult neurogenesis in the SVZ and DG of mammals, their structural characteristics, function, lineages, and signaling pathways have, for obvious reasons, has been dominated by the classical model organisms – mice and rats. But we may learn much from examining other species – structural and seasonal variations, age-related declines in neurogenesis, methods to increase proliferation or neurogenesis, ecological demands on neurogenesis, the role of DG neurogenesis in spatial learning and memory – and to look at these aspects a range of other mammals have been studied including humans, non-human primates, cows, rabbits, gerbils, guinea pigs, bats, cetaceans, and moles (Amrein et al., 2007; Eriksson et al., 1998; Gil-Perotin et al., 2009; Iwai et al., 2003; Jabès et al., 2010; Kuhn et al., 1996; Parolisi et al., 2015; Ponti et al., 2006; Quiñones-Hinojosa et al., 2006; Rodríguez-Pérez et al., 2003). At

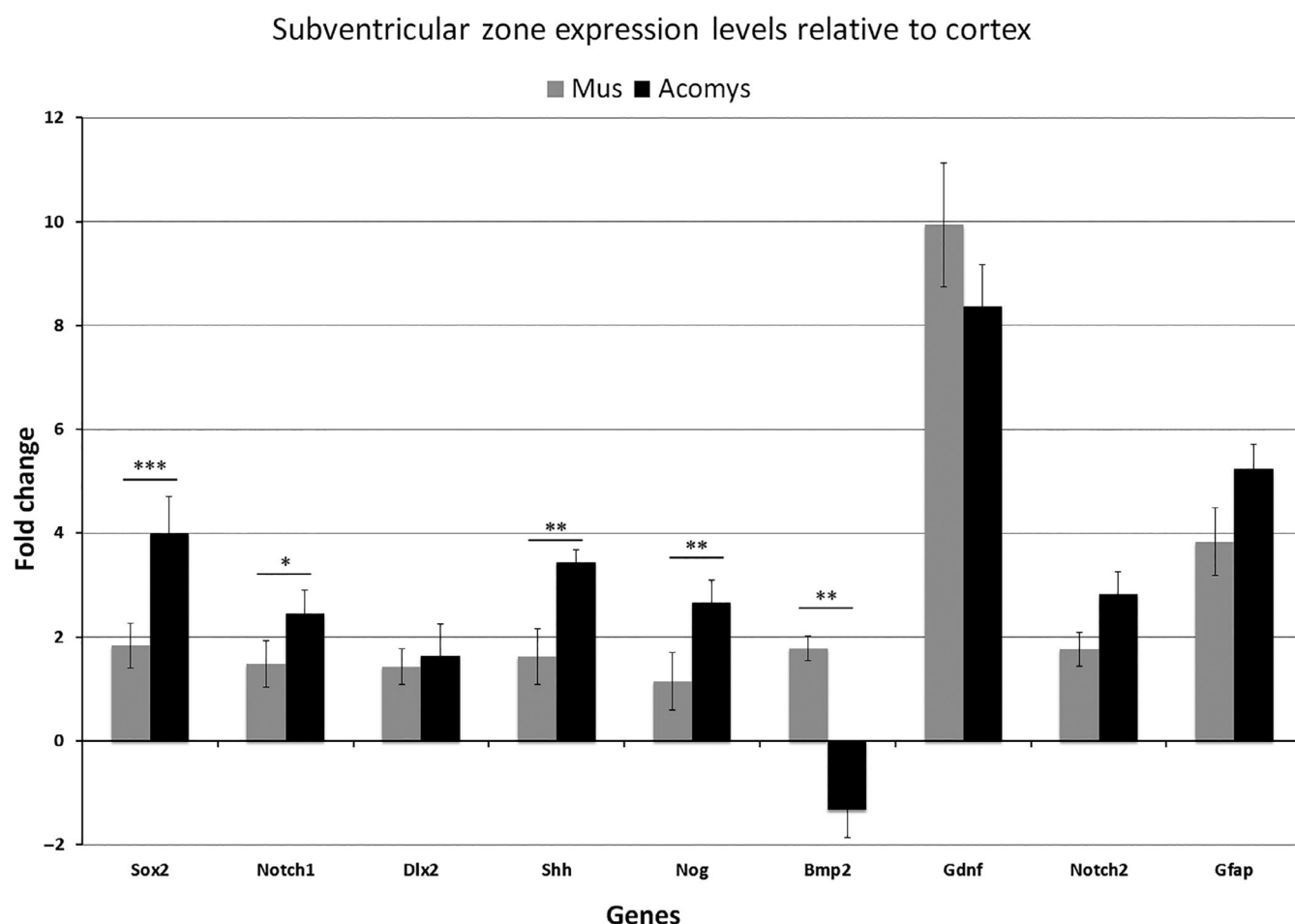


FIGURE 6 RT-qPCR data for nine genes associated with the subventricular zone (SVZ) niche. The data is expressed as the relative levels in the SVZ compared to the levels of the same gene in the cortex. Grey bars *Mus*, black bars *Acomys*. Except for *Bmp2* in *Acomys* all genes are expressed more highly in the SVZ than in the cortex, but only *Sox2*, *Notch1*, *Shh*, and *Noggin* are significantly expressed more highly in *Acomys* than in *Mus*. *Bmp2* is expressed significantly lower in *Acomys* than in *Mus*. The remaining genes show no significant difference. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

least in the DG the lowest level of expression is seen in humans, nonhuman primates, bats, cetaceans, shrews and moles (Amrein, 2015; Lipp & Bonfanti, 2016) and at the other end of the scale rodents appear to have higher levels of adult neurogenesis than other mammals. This variation does not seem to fit with the crucial role for DG neurogenesis in learning and memory since those species which might be expected to have the highest need for this property such as primates, have the lowest levels of neurogenesis (Amrein & Lipp, 2009). Another example is bats which have no neurogenesis in the DG despite showing precise spatial memory for food sources in the wild (Amrein et al., 2007).

Nevertheless within groups there are some correlations between proliferation, neurogenesis and spatial abilities. For example there were four times as many proliferating cells in the DG in two wood mouse species than in bank and pine voles with laboratory mice showing rates in between the two (Amrein et al., 2004). So there may be some significance in the fact that wood mice patrol large territories and have these high rates of proliferation (Amrein & Lipp 2009) and have memory retention superior to laboratory mouse strains (Patil et al., 2008). Similarly, the number of DCX-positive neurons and double-labeled DCX-BrdU-positive neurons correlated with performance on several cognitive tasks including spatial memory and discrimination learning in dogs (Siwak-Tapp et al., 2008).

Whatever the case may be, it is a useful exercise to examine NSCs in many different species and here we have examine them in *Acomys cahirinus*, the spiny mouse. Our rationale for this is that this unique mammal has very significant regenerative properties in the skin, ear, skeletal muscle, heart, spinal cord, and kidney (see Section 1). Furthermore, the brain of these animals does not respond to the implantation of a foreign body by forming a glial encapsulation suggesting a differential response to damage is also present in the CNS, as we have shown for the spinal cord (Streeter et al., 2020). We have found here that there is a very significant increase in the number of NSCs of all types (A cells, B cells, and C cells) in the SVZ and RMS of *Acomys* such that the walls of the lateral ventricle are greatly thickened as is the size of the RMS. In the horizontal wall of the SVZ, there were three times as many cells as compared to laboratory mice, three times as many Ki67+ve proliferating cells and 3.6 times as many cells which had taken up BrdU.

Similarly, in the other site of NSC production, the dentate gyrus, in *Acomys* there were 5–6 fold more BrdU+ve cells, 11–12 fold more Ki67+ve cells, 11–12 fold more PNCA+ve cells and 3–4 fold more doublecortin+ve cells. These results suggest that at both sites of NSC production *Acomys* is producing many more cells and many more neuroblasts for incorporation into the olfactory bulbs and dentate gyrus for potentially improving the functioning of these physiological units.

It is possible that these increased cell numbers in the SVZ and DG are simply due to more cells in the larger *Acomys* brain which weighed 50% more than the lab mouse brain because *Acomys* were larger animals. Their brain:body weight ratios were, however, the same at 1.4%. Also arguing against this larger brain:more NSCs hypothesis is the data we obtained from wild caught *Mus musculus*

domesticus which had far larger brains that either *Acomys* or lab mice (50% higher brain weights) despite being smaller animals (40% lower body weights than lab mice) giving a brain:body weight ratio of 7.5%, about 5–6× the figure for *Acomys* and lab mice. Even though the wild caught *Mus* had far larger brains than lab mice they still had a *Mus*-like proliferation rate in the DG of 12 Ki67+ve cells per section compared to 5 cells per section in *Mus* and 54 cells per section in *Acomys*.

Another confounding issue could be age since the rate of NSC production decreases dramatically with age (Amrein et al., 2011; Kuhn et al., 1996) so the age of the species to be compared should be carefully considered. Since *Acomys* live twice as long as lab *Mus* we chose to use *Acomys* animals that were twice as old (6 months *Acomys*, 3 months *Mus*) for the comparison. Even so, *Acomys* had far higher NSC numbers and proliferating cells and if this was simply due to age then the numbers in *Acomys* would be expected to be decreased, not increased.

We began a preliminary analysis of gene expression differences between the SVZ in *Mus* and *Acomys* using a selection of signaling genes, growth factors and transcription factors which are known to play a role in promoting NSC proliferation and differentiation (Ehm et al., 2010; Lai et al., 2003; Yousef et al., 2015). We examined *Sox2*, *Notch1*, *Dlx2*, *Shh*, *Noggin*, *Bmp2*, *Gdnf*, *Notch2*, and *Gfap* and showed that in both *Mus* and *Acomys*, eight of these genes were indeed enriched in the SVZ compared to the cortex, the only exception being *Bmp2* which was not enriched only in the *Acomys* SVZ, rather it showed lower levels. Furthermore, in the case of *Sox2*, *Notch1*, *Shh* and *Noggin* the level of enrichment was significantly higher in *Acomys* than in *Mus* which may suggest that these genes contribute in a positive manner to the greatly increased number of NSCs seen in the SVZ of *Acomys*. Indeed, *Sox2* is the neural precursor marker and higher levels of this may reflect higher absolute numbers of NPCs in *Acomys*. *Notch1* signaling maintains proliferative and undifferentiated stages of NPCs and its overexpression promotes proliferation, at least in the DG (Breunig et al., 2007). *Shh* overexpression promotes proliferation in the DG (Lai et al., 2003). *Noggin* is a negative regulator of BMP and increased levels of *Noggin* could lower BMP levels. Since BMP levels increase during aging and their attenuation leads to increased NSC proliferation (Yousef et al., 2015) lower levels of this growth factor in *Acomys* could lead to the increased numbers of proliferating cells that we have observed. Thus, all of these genes changes we noted could feed into increased NPC proliferation in *Acomys*.

What is likely to be the significance of these increased numbers of NSCs in *Acomys*? It is possible that *Acomys* has better olfactory discrimination than the lab mouse due to the increased number of cells in the RMS. In the DG, it is possible that *Acomys* have better exploratory abilities or spatial memory. It is unlikely to be the case that inbred laboratory animals have evolved lower NSC production in the DG because our wild caught *Mus* have similar production levels to our lab *Mus*. Future studies may investigate these possibilities, but from the regenerative potential it may be that this significantly increased NSC production levels in *Acomys* coupled with a decreased cytokine response to damage

permits a better recovery potential after damage to the brain such as that received after a stroke. Future experiments should be directed towards this goal.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

AUTHORS' CONTRIBUTIONS

MM designed and performed the experiments and wrote the paper. NS performed the histology, immunocytochemistry, and cell counts. MB performed the histology and immunocytochemistry. AGWS performed the PCR.

DATA AVAILABILITY STATEMENT

Data available on request from the authors.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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